

What Do Impact Investors Do Differently?

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Do impact investors seek impact, or merely “impact wash”? We provide systematic evidence on the nonfinancial determinants of impact investing. Impact investors focus on firms aligned with the priorities of the federal government and disproportionately invest in economically disadvantaged regions. While we find high levels of coinvestment between impact and traditional investors, we also show that impact investors influence the strategies of their traditional coinvestors, are more likely to fund firms in nascent industries, and invest countercyclically. Finally, we characterize investment heterogeneity based on a novel classification of impact investment theses, with a focus on climate, environment, and jobs and equity. (JEL G11, G23, G24, H41, M14)

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In recent years, impact investors, private fund managers who seek to generate both financial and social returns, have attracted intense interest. Established alternative investment groups—such as Bain Capital, KKR, and TPG—have raised substantial funds seeking to fulfill these twin goals. Meanwhile, dedicated impact-focused groups have proliferated. Estimates suggest that worldwide, private debt and equity impact investors manage more than \$850 billion (Hand et al. 2024a, b).¹

This boom in activity has proven controversial. Industry proponents have argued that the substantial capital controlled by impact investors will hasten the achievement of the United Nations (UN) Sustainable Development Goals (SDGs).² UN Secretary-General António Guterres is representative of the advocates for this view: “Investing in the SDGs is both sensible and feasible: it is a win-win.”³ Such investments might be particularly valuable in an era with diminished public social funding in many developed nations.

On the other hand, many industry observers remain skeptical. One line of criticism accuses impact investors of “impact washing”—that is, deploying their funds in similar portfolios to those of traditional investors (see, e.g., [Cote 2022](#)). For instance, Tariq Fancy, BlackRock’s former Chief Investment Officer for Sustainable Investing, argued that “sustainable investing boils down to little more than marketing hype, PR spin and disingenuous promises from the investment community” (Fancy 2021, n.p.). A related critique maintains that even if impact investors prioritize social and environmental impact in their portfolios, they will not advance social impact if they focus on companies that traditional venture investors would also have funded. Put another way, impact investors must be “additional” to have an impact ([Brest and Born 2013](#)).

While there has been some academic research on the financial returns of impact investors ([Kovner and Lerner 2015](#); [Barber et al. 2021](#); [Jeffers et al. 2024](#)), much less work has examined the social and environmental impact of impact investing. Two notable exceptions are [Geczy et al. \(2021\)](#), which studies how contracting choices in impact investing relate to measures of social impact, and [Lerner et al. \(2025\)](#), which documents that companies

¹ This estimate combines the estimate reported in Hand et al. (2024a) that impact investors manage upwards of \$1.5 trillion worldwide with the estimate reported in Hand et al. (2024b) that 57% of all capital in the impact investing market is controlled by private debt and equity impact investors.

² For instance, at the time of this writing, the UN SDGs are prominently mentioned on the websites of ResponsAbility, Calvert Impact Capital, and LeapFrog Investments—three prominent impact funds included in our data set.

³ For the press release in which this quote was reported, see <https://news.un.org/en/story/2023/02/1133637>

are more likely to hire and promote minority and unskilled workers after receiving impact financing.

This paper seeks to characterize the nonfinancial determinants of impact investment to shed light on the debate outlined above. Our analysis proceeds along three lines. First, we examine several proxies for the social impact of impact investors to investigate the degree to which impact investors follow differentiated strategies, versus merely “impact wash.” Second, investigating the [Brest and Born \(2013\)](#) additionality critique, we document how impact investors cooperate and compete with traditional venture investors, sometimes providing capital that substitutes for traditional venture investment and sometimes providing capital to ventures that would have been overlooked by traditional venture investors. Finally, we revisit both of these questions with an eye toward heterogeneity among impact investors, utilizing a novel classification of investors based on the characteristics of their portfolios.

To perform these analyses, we construct a new data set covering a broad spectrum of impact investors. We identify impact investors by combining a wide variety of data from impact organizations, investment group websites, and commercial databases.⁴ To compare the investment activity of impact and non-impact-investors, we use activity as recorded in PitchBook.

We employ several approaches to investigate proxies for the direct social impact of impact investors. First, we document that impact investors finance companies aligned with the priorities of the U.S. federal government, as proxied by receipt of a Small Business Innovation Research (SBIR) award and receipt of a patent with a noted government interest, including patents where the government was an assignee but also those that were government licensed or funded ([Gross and Sampat 2024](#)). Relative to those of traditional venture investors, impact investors’ portfolio companies are 80 to 180% more likely to have these. Furthermore, we find suggestive evidence that impact investors not only select portfolio companies aligned with the priorities of the federal government but also that they causally shift portfolio companies toward this alignment. Utilizing a difference-in-differences framework, we estimate that compared to a traditional venture investment, an impact investment nearly doubles the likelihood of receiving an SBIR award (though we find no similar impact on the likelihood of filing a patent with a government interest). This finding suggests a channel through which impact investors may be “additional”—the presence of impact investors in the capital structure of the portfolio company meaningfully increases the probability the company pursues an SBIR award.

⁴ We will make these data on impact investors available to interested researchers.

Our final proxy for social impact comes from the characteristics of the location in which a portfolio company operates. We explore a battery of metrics related to whether an area is economically disadvantaged. At the international level, impact investors finance companies from countries with a per capita GDP of about \$4,000 less than those financed by traditional investors. Within the United States, we do not find substantial evidence that, on average, impact investors finance companies in more disadvantaged areas. However, as we elucidate later, this result masks substantial heterogeneity across different types of impact investors.

Our next line of inquiry explores the interaction between impact and traditional investors. To what degree do impact investors provide novel forms of financing that reach firms that may have been missed by the traditional venture capital (VC) and growth equity markets? A first signal comes from the extent of coinvestment between traditional and impact investors—that is, investors providing financing to the same company in the same round of fundraising. Perhaps surprisingly, despite having distinct investment theses, impact investors coinvest with traditional venture investors in about 60% of their investments, nearly twice the rate at which traditional investors coinvest with other traditional investors. This suggests that a majority of impact investments are attractive to traditional venture investors as well.

We also find evidence that impact investors influence the behavior of some traditional investors. We examine the investment behavior of traditional investors that have at one point coinvested with an impact investor. We find that after an impact investor raises a fund, the subsequent investments of their prior coinvestors are 7.4 to 17.4% more likely to have an SBIR award or government-interested patent, indicating increased alignment with the priorities of the federal government and, as per our prior results, the priorities of impact investors. Fundraising by an impact investor could affect its previous coinvestors in at least two ways. First, it could increase the probability that traditional investors directly coinvest with an impact investor that recently raised a fund. Second, there could be a “demonstration” effect, whereby impact investors influence their previous coinvestors to favor specific impactful sectors without directly coinvesting (e.g., through discussions and inference about promising investment opportunities). Our estimates suggest this second channel is the most critical.

We further identify two ways in which impact investors provide capital that complements traditional venture investment. First, impact investors are substantially more likely to invest in nascent industries whose financial viability may not be sufficiently established for traditional investors. This narrative is sometimes used to justify impact investing (e.g., to support the creation of new industries like microfinance), but to our knowledge, there has not been systematic evidence demonstrating that impact investors fill

this role. We find that impact investors' portfolio companies are about 50% more likely to be among the earliest to raise capital in their PitchBook industry compared to the portfolio companies of traditional investors.

Second, we demonstrate that impact investment is countercyclical relative to market valuations. It is well established that VC is strongly procyclical (Gompers et al. 2008). We confirm that traditional venture financing is strongly correlated with the market-to-book ratio at the industry sector-by-year level. Critically, however, we find that a sector's market-to-book ratio is somewhat negatively predictive of the volume of the sector's impact investment, indicating that impact investment is countercyclical. This is consistent with the possibility that there may be greater opportunities for social impact when traditional venture financing is constrained.

Finally, we revisit both lines of inquiry described above with an eye toward heterogeneity among impact investors. We classify impact investors as falling into three broad categories—Climate, Environment, and Jobs and Equity—based on whether at least 20% of their investments can be classified as such. While most of our findings generalize across the different types of impact investors, several forms of heterogeneity emerge. First, while the impact investing industry as a whole does not invest in poorer geographies within the United States, impact investors focused on Jobs and Equity do focus on disadvantaged regions, while the reverse is true for those focused on Climate and the Environment. Correspondingly, the portfolio companies of impact investors focused on Jobs and Equity, and to a lesser degree those of impact investors focused on the Environment, are less likely to reach an IPO, merger, or acquisition.

We also find that Climate and Environment specialists are most likely to have a patent with a government interest; this result does not hold for concessionary impact investors or for those specializing in Jobs and Equity. Concessionary impact investors and those specializing in Jobs and Equity seem to primarily invest in companies for which formal intellectual property is less important. Similarly, we find that while, relative to traditional investors, all impact investors are more likely to invest in companies belonging to nascent industries, this proclivity is substantially stronger for impact investors focused on Climate and the Environment.

Our analysis suggests a few takeaways. First, we highlight that impact investors indeed fund high-impact firms, indicating they are not merely “impact washing.” Second, we document the ways in which impact investors influence their traditional coinvestors, and provide capital in nascent industries and during downturns when traditional investors are unwilling to invest—that is, that impact investors are, at least sometimes, “additional.” Finally, we demonstrate the importance of heterogeneity among impact investors by providing a novel classification of their investment focuses and demonstrating substantial differentiation.

1. Data and Descriptive Statistics

1.1 Data construction

This paper is the first to use a newly created data set that we believe is the most comprehensive data set on impact investors and their portfolio companies. The data cover the period from 1980 to 2021. An accompanying technical paper ([Burton et al. 2021](#)) describes the data construction process in detail. Here, we focus on the key elements. [Internet Appendix A](#) provides more details on the data set construction.

We define impact investors as investors with the explicit dual objective of generating social good and financial returns.⁵ To compile our catalog of impact investors and portfolio companies, we draw on information in multiple financial databases, performing extensive matching and data quality checks. We then compare our results with expert judgments, published reports, and other independent research to remove investors that do not target both social and financial returns.

We identify potential impact investors using nine established resources on impact investing:⁶ (1) ImpactBase, the global directory of impact investment funds from the Global Impact Investing Network (GIIN); (2) the Community Development Venture Capital Association (CDVCA) website; (3) the Impact Assets website; (4) Preqin's alternative assets database; (5) members of Impact Capital Managers (ICM), a consortium of impact investors; (6) the list of asset managers who are GIIN members; (7) GIIN's Investors' Council members; (8) the signatories to the Operating Principles for Impact Management originated by the International Finance Corporation; and (9) the Private Equity International (PEI) "Impact Investment Firm of the Year" top three honorees.

Aside from Preqin, all of these are special compilations that focus specifically on impact investors. In Preqin, the "fund ethos" variable allows investors to self-identify as having a focus on at least one of the following five categories: "Microfinance," "Economic Development," "Socially Responsible," "Environmentally Responsible," and "Sharia Compliant." We expand this preliminary list by adding investment firms whose stated industry focus corresponds with so-called impact sectors. In particular, we add investment firms that primarily invest in "Clean Technology," "Education/Training," and "Environmental Services." Finally, we add

⁵ There is not yet a single widely adopted definition of impact investing; we discuss our definition and how it relates to the literature in the [Internet Appendix](#).

⁶ The versions of the databases that we used were as follows: ImpactBase as of January 15, 2018; Community Development Venture Capital Association (CDVCA) as of May 2019; Impact Assets for the period 2011–2019; Preqin's alternative assets database as of June 30, 2018; Impact Capital Managers (ICM) members as of May 2020; Global Impact Investing Network (GIIN) members as of May 2020; GIIN's Investors' Council members as of May 2020; signatories to the Operating Principles for Impact Management originated by the International Finance Corporation as of May 2020; and the Private Equity International "Impact Investment Firm of the Year" top three honorees for the years 2017–2019.

investment firms that primarily invest in low-income countries, identified as those countries with a GDP per capita of less than U.S. \$1,400.⁷ This process results in a total of 2,747 potential impact investors for further investigation.

We then narrow this set by eliminating those that do not align with our definition of impact investors. We manually search their websites, if available, to see if they make any mention of a dual aim of generating social and financial returns.⁸ In addition, we eliminate development finance institutions such as the International Finance Corporation (a subsidiary of the World Bank). We combine the information from all of the above listed sources to create a list of 445 unique impact investors, which includes some standalone impact funds and traditional private equity (PE) firms that have large impact investment funds.⁹

In [Internet Appendix Table A.I](#), we summarize how we create our final set of impact investors used in our analyses. First, we review the 445 impact investors' websites and exclude 46 impact investors whose strategies do not focus exclusively on impact investing and groups that were launched without an impact mandate but subsequently added one. Here, we looked for language that was more specific than "do good" or "make the world a better place." We included all investors that articulate a goal of promoting economic growth in a specific region, alleviating poverty, or benefiting disadvantaged individuals. However, investors with a focus on specific industries (EdTech or Health Care, for example) were not automatically categorized as impact, unless they articulated a social mission. For instance, we only include investors in biotech firms that have a target objective beyond the financial returns in the development of a drug, such as helping disadvantaged persons gain access to life-saving medication. Lastly, we eliminate three foundations.

⁷ World Bank estimates of the typical poverty line in middle-income developing nations—which are more likely to be the focus of impact investors than the very poorest nations—range between \$3.65 and \$4.00 per day (in PPP-adjusted dollars). See, for instance, <https://www.worldbank.org/en/topic/poverty/brief/global-poverty-line-faq#:~:> and <https://pipmaps.worldbank.org/en/data/datatopics/poverty-portal/poverty-geospatial?dataset=PovertyRate2.15-gsap&zoomLevel=3&lat=19.53676432208408&lng=15.0234375000001>.

⁸ We accomplish this by using Amazon's crowdsourcing marketplace, Mechanical Turk ("MTurk"), and its online workforce of "MTurkers." We asked the MTurkers to collect the description, stated mission, and investment strategy as listed on the potential impact investor's website and to identify whether they make mention of the dual aim of generating both financial and social returns. For each potential impact investor, we asked three MTurkers to review its website. If two of three MTurkers voted to exclude an investor, it was excluded. Using this approach, we narrow the list of 2,747 to 624 potential impact investors. Again, following [Barber et al. \(2021\)](#), the remaining 624 were then manually verified by a member of the Project on Impact Investments team through a careful review of the background and strategy on the impact investor's website to identify any mention of the dual objectives of social impact and financial returns. Only those investment managers who make explicit statements that signal a dual objective were classified as impact investors.

⁹ Through this process, we identify 199 impact investors from Preqin, compared to the 159 identified by [Barber et al. \(2021\)](#) in the period from 1995 to 2014.

Our initial screen of investors solely identified standalone impact investors. However, this screening would have eliminated a set of 13 investors we thought appropriate to include: traditional PE firms with significant funds dedicated to impact investing. We identified these dedicated funds through the extensive media coverage of impact investing. Our research uncovered four funds with investments in the PitchBook database: Bain Double Impact, TPG Alternative & Renewable Technologies, The Rise Fund (also a TPG affiliate), and PG Impact Investments (an affiliate of Partners Group). The other groups, such as Black Rock and KKR, had raised capital for impact funds in 2020 or 2021 but had no transactions by these units in the PitchBook database as of the May 2021 version of the database we used.¹⁰ We are left with 396 impact investors.

We match the 396 impact investors to the May 2021 PitchBook universes of preventure, VC, growth equity, and PE investors (203,898 entities). Of the 396 impact investors, 291 match to the PitchBook data feed.¹¹ We drop one investor because it has no deal information and nine investors because they do not have any VC or growth equity investments. Finally, after additional data cleaning comprised of removing subsidiaries (with the exception of the dedicated impact groups of generalist organizations) and groups with only failed transactions, we are left with 277 impact investors in our study (two of which are affiliated with the same general partner, TPG). [Internet Appendix Table A. III](#) provides a complete list of the 277 impact investors included in the analysis.

Having created this list of impact firms, we wish to compare their investment activity to other private investing groups. The source of our data on portfolio companies of both impact and traditional investors is PitchBook, one of the most comprehensive databases linking investors to their investments. We did not use any data set that lists only impact-specific investments, as we wanted an equivalent level of comprehensiveness for both impact and traditional firms. We detail our sample inclusion criteria in [Internet Appendix A](#); the following paragraphs provide an overview.

To create a comparable set of traditional non-impact-investors, we use the PitchBook data feed as of May 2021 and extract all investors in the VC and PE universes, identifying over 200,000 investors. We remove the 445 impact investors as well as investors assigned to categories which do not have VC or growth equity as a main part of their overall investment strategy.¹² Of the remaining investors, we further restrict our sample

¹⁰ Some absences may reflect difficulties in identifying transactions made by the impact affiliates of these large groups, rather than those by their main funds.

¹¹ There are 291 general partner-level investors, one of which has two separate funds (TPG).

¹² We exclude the following PitchBook categories of investors: Angel (individual), Business Development Company, Corporate Development, Corporate Venture Capital, Corporation, Family Office, Fund of Funds, Fundless Sponsor, Government, Hedge Fund, Holding Company, Investment Bank, Limited

to focus on investors that have at least four private capital portfolio companies, thus removing investors that may only have one-off VC or growth equity investments (e.g., we do not want to include a mutual fund that has a few PE investments, where PE is not a main part of its investment strategy).^{13,14} In addition, as in the case of the impact investors, we drop groups with only failed deals and those with no venture or growth equity rounds in the PitchBook database.¹⁵

For the portfolio companies of both impact and non-impact-investors, we eliminate duplicate entries and firms in which all financing rounds have missing information (in particular, the date of investment, the number and identity of coinvestors, and the total size of each round of funding).

At the end of the process, the 277 impact investors in the database have made investments in a total of 6,066 portfolio companies. The comparable set of non-impact-investors includes about 20,000 traditional investors, which have invested in approximately 205,000 companies.

To conduct our study of the demographic determinants of impact investments in [Section 2](#), we match our portfolio companies with the census demographic data. To do this, we first match the portfolio companies' zip codes to the Federal Information Processing System (FIPS) codes that uniquely identify counties. The match rate between the zip codes

Partner, Merchant Banking Firm, Mutual Fund, Other, Private Equity-Backed Company, Secondary Buyer, Sovereign Wealth Fund, Special Purpose Acquisition Company, University, and Venture Capital-Backed Company. We do this to maintain comparability to our sample of impact investors. The fact that we exclude operating companies undertaking roll-ups and other consolidation strategies reflects the fact that impact investors overwhelmingly target young startup companies or else already-profitable companies with high growth potential: that is, they focus on funding innovation, product development, and scaling. While there are also VC/PE-backed companies that behave like PE investors engaged in consolidations or roll-ups, the emphasis on cost-cutting in these transaction is very different, as seen for instance on the evidence on the employment impact of different classes of transactions (see, for instance, [Boucly and Thesmar \(2011\)](#); Puri and Zarutskie (n.d.); [Davis et al. \(2014\)](#)). To maintain comparability, we exclude such transactions from the sample of comparable transactions.

- ¹³ Given the comparatively smaller number of impact investors in our data set, we were able to manually assess each investor to ensure that VC and growth equity investing are a core part of its strategy. Thus, we do not apply the criterion that impact investors have at least four VC or PE deals. Such an effort was not really feasible for the tens of thousands of small non-impact-investors. In many cases, family offices have names that resemble those of formal investment groups. Efforts to research family offices with little investment activity frequently are fruitless, as they typically attract little media attention and aggressively use Delaware and Cayman Island shell companies to obscure their ownership. For a discussion of these issues in the context of Chinese family offices, see [Akcigit et al. \(2024\)](#).
- ¹⁴ In unreported analyses, we verify the robustness of our results to applying a five-portfolio company threshold for our matching traditional venture investors. We have also verified the robustness of our results to the three-portfolio company cutoff, with the exception of [Table 7](#), which would require matching additional firms to their fundraising data, and our heterogeneity analyses based on investor thesis ([Table 11](#)), which would require classifying additional investors.
- ¹⁵ This exclusion affects a very small share of the sample: 2.75% (572 investors) of traditional investors and 1.1% (5 investors) of impact investors. Further, the majority of our analyses rely on completed deals, which would regardless exclude investors with only failed deals.

and the FIPS codes is 94.4%. Next, we match the portfolio companies' FIPS codes and deal year to the census data. Here the match rate is 84.3%.¹⁶

For the analyses in Section 2 examining the overlap of impact investments with government priorities, we also match the portfolio companies with SBIR awards and U.S. government-interested patents. We use the SBIR database at www.sbir.gov/awards#, which included in the version we accessed (March 27, 2024) all awards from the program's inception in 1983 to the end of 2023, and the database of government-interested patents compiled by Gross and Sampat (2024), which includes government-interested patents that they were able to identify since the inception of the modern patent system in 1836 through the end of 2020.

We performed this match in a three-step process. We started with the subset of U.S.-based impact-backed companies and those backed by traditional investors with a location and company name. (Eligibility requirements for the SBIR program, for instance, excluded foreign firms.) First, we matched portfolio companies where there was an exact match based on city and company name in the PitchBook and the SBIR databases. Second, we did an exact match based on state and company name. Finally, we did an exact match based on state and a fuzzy match based on company name, manually checking all of the awards. We matched 8.7% of the impact portfolio companies and 5.5% of the non-impact-portfolio companies to the SBIR database, reflecting the fact that most companies did not have such awards. We performed a similar matching process on patents. In total, 24.5% of the impact portfolio companies and 16.9% of the non-impact-portfolio companies were matched to patents, whether ones with a government interest or not.¹⁷

1.2 Classifying the impact investors

To examine heterogeneity across impact investors, we focus on two dimensions: the type of UN SDG that they address and whether they are concessionary.

To assign investors to what we term "SDG buckets," we follow the methodology delineated in Internet Appendix A.4. We use the 17 SDGs adopted by the UN in 2015 with the goal of ending poverty, protecting the planet, and ensuring peace and prosperity. We identified a set of keywords associated with each SDG. We then had three research associates

¹⁶ Prior to 2009, the census demographic data are only available each decade. After that, the data are available on an annual basis. Thus, we have data for the years 1999 and 2009–2020. We match deals made in the years from 1990 to 1999 to the census data from 1999 and deals made in the years from 2000 to 2009 to the census data from 2009.

¹⁷ We manually checked the awards to non-impact-firms only if the similarity score from the fuzzy match was 96 or above; below this, there was a sharp drop-off of match quality, so we dropped these potential matches. For the impact firms, all potential patent matches above 90 were checked; we automatically dropped all matches with a score below 90.

classify 6,000 portfolio companies in two rounds, using the aforementioned seeding words as guidance. We then subjectively consolidated the 17 categories into four buckets: “Climate,” “Environment,” “Jobs and Equity,” and “Social Infrastructure.”¹⁸ With the resulting training data, we fine-tuned a Bidirectional Encoder Representations from Transformers (BERT) model to score PitchBook descriptions of each portfolio company against the five “buckets”—the four SDG buckets and a “No bucket” category (i.e., not an impact company).

Having used the text to classify portfolio companies into SDG buckets, we then classified impact investors. We first counted the total number of deals for each impact investor over the entire sample period and the number that fell into the four SDG buckets for each investor. As there were only 44 deals in the social infrastructure bucket, made by only 8 investors, we omitted it from our subsequent analysis, and considered only the three other SDG buckets.

For each impact investor, if 20% or more of the investors’ deals were in a given SDG bucket, we associated the impact investor with that bucket. Of the 277 impact investors, 167 are recorded as falling into only one of these 3 buckets. Fifty-three are assigned to two such focuses, and two have three.

We note that PitchBook descriptions may have been updated after an investor provided financing to a portfolio company. Thus, for any given company, we cannot be sure whether its classification in a bucket is a “treatment effect” of an investor or whether its classification was unaffected by its investors. However, since our goal is to classify impact investors based on whether their portfolio companies fall in a given SDG bucket, we are comfortable remaining agnostic about the timing of when their portfolio companies received their classification.

[Internet Appendix B](#) provides examples of nine companies that fall into the three buckets, in each case two that match with high confidence and one with lower confidence.

We also identify concessionary impact investment firms: those that self-identify that they target or accept below market-rate returns. For the impact investors in the sample, we examined their websites for language related to their target return objectives. We classified firms into three categories:

- **Concessionary:** If we found “below-market,” “concessionary,” “social returns,” “impact returns,” or similar terms with no mention of achieving market-level financial returns, we classified them as concessionary. An example would be, “We believe in making economically rewarding investments while focusing primarily on tangible social and environmental benefits.”

¹⁸ Though the SDGs were adopted in 2015, the four groups we utilize to characterize them also appropriately characterize all but one of the Millennium Development Goals that preceded them.

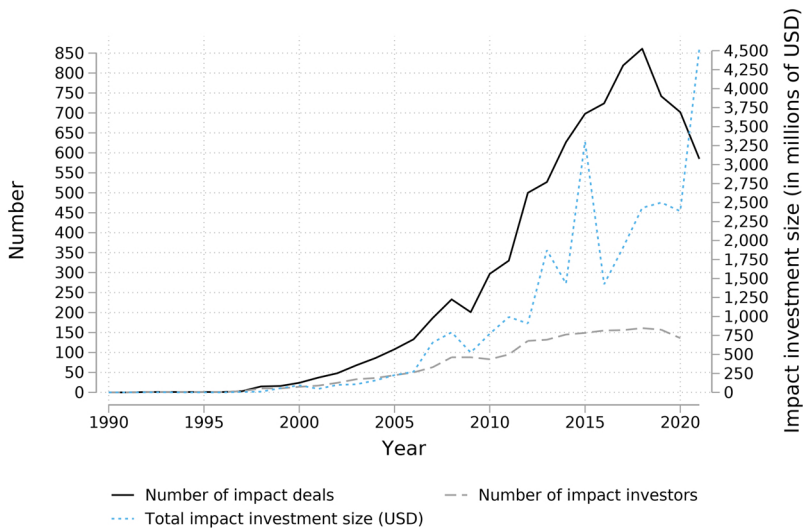


Figure 1
Number of impact deals, number of impact investors, and estimated investment amount by year
In this figure, we plot the total number of impact deals (left axis), number of unique impact investors (left axis), and total investment from across impact deals in millions of USD (right axis) in our data set between the years 1990 and 2021. We have data only from January to April 2021, so we normalize the 2021 number of impact deals and total impact investment size by linearly projecting based on the first four months in 2021. There were no impact deals in the years 1990 and 1991. We do not observe the investor-specific financing in each round; we only observe the total financing by all investors in that round. We therefore divide the total financing in each round by the number of investors in that round and assign that (equal) portion to each investor. The outcome in the right axis is thus the sum of this measure for all impact investors in each year. During the sample period, a total of 277 impact investors invested in 8,125 unique deals involving 6,066 distinct portfolio companies.

- **Market return:** Here we look for mention of achieving market or above-market financial rates of return.
- **No return information:** Firms that did not mention achieving any type of return—financial or social returns—in their objective.

Of the 277 impact investors, 47 were classified as concessionary.

1.3 Descriptive statistics

In Figure 1, we document the growth of the impact investment sector, plotting over time the number of impact deals, the number of impact investors, and an estimate of the total dollar value of impact financing. Figure 2 plots the number of impact investors and the total dollar value of impact financing broken down by SDG buckets over time. Thirty years after the birth of impact investing, we see that 6,066 firms have received funding from impact investors, in 8,125 investment rounds. These represent approximately 2% of all VC and growth equity rounds and 3% of all VC and growth equity enterprises.

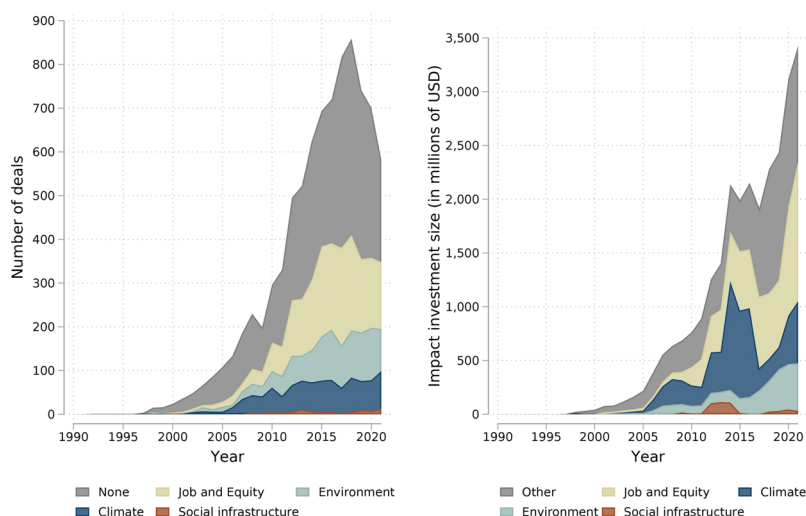


Figure 2

Number of impact deals and moving average of investment amount by SDG bucket and year

In this figure, we plot the total number of impact deals (left graph) and a moving 3-year average of the total investments from across impact deals in millions of USD (right graph) in our data set between the years 1990 and 2021 across four SDG buckets. We have data only from January to April 2021, so we normalize the 2021 number of impact deals and total investment size by linearly projecting based on the first 4 months in 2021. The moving average is calculated by averaging the focal year with 1 year prior and after. For 2021, we average only the 1 prior and the focal year (2021). There were no impact deals in the years 1990 and 1991. We do not observe the investor-specific financing in each round; we only observe the total financing by all investors in that round. We therefore divide the total financing in each round by the number of investors in that round and assign that (equal) portion to each investor. The outcome in the right graph is thus the sum of this measure for all impact investors in each year in the SDG bucket.

The aggregate totals in [Figure 2](#) mirror those in [Figure 1](#), with the exception that we present the 3-year moving average for investment amounts to smooth some volatility introduced by large individual transactions. Jobs and Equity is the single largest category. The figure also makes clear the intense cyclicity of the dollar volume of Climate investing, with booms in the late 2000s, early 2010s, and 2020s, followed by busts.

[Table 1](#) provides basic descriptive statistics about the investors in our sample. Column 1 presents the counts and means for the portfolios of traditional venture and growth investors and the companies they back. Column 2 presents the difference between these groups and impact investors for each outcome. We defer the discussion of the remaining columns until [Section 4](#).

Panel A of [Table 1](#) shows that on average, traditional investors and impact investors have similar portfolio sizes, with the former having supported an average of approximately 24 companies with 31 investments, and the latter slightly more on each measure. The average traditional and the average impact investor have been in operation for approximately 10 years. However, there are substantial differences in average deal size: the

Table 1
Summary statistics by investor type

	(1) Traditional investor mean	(2) Impact investor difference	(3) Climate impact investor difference	(4) Environment impact investor difference	(5) Job and Equity impact investor difference	(6) Concessionary impact investor difference
<i>A. Portfolio profile</i>						
Number of companies	24.438	1.042 (3.637)	-8.453*** (1.998)	-7.549*** (2.057)	5.232 (7.335)	-5.161 (4.532)
Number of deals	30.612	1.399 (4.086)	-8.744*** (2.922)	-9.312*** (2.547)	5.578 (7.853)	-7.719 (5.326)
Average investment size	8.703	-3.720*** (0.754)	-1.196 (1.724)	-3.534*** (0.854)	-4.521*** (0.915)	-4.620*** (1.115)
Years in operation	9.902	-0.197 (0.310)	-0.444 (0.540)	-0.345 (0.474)	-1.183*** (0.407)	-0.289 (0.771)
<i>B. Global regions</i>						
US	0.451	-0.007 (0.026)	-0.001 (0.049)	-0.009 (0.043)	-0.161*** (0.035)	-0.026 (0.063)
Advanced economies	0.433	-0.240*** (0.019)	-0.109** (0.045)	-0.265*** (0.028)	-0.254*** (0.027)	-0.221*** (0.048)
Developing economies	0.112	0.250*** (0.025)	0.112*** (0.042)	0.278*** (0.044)	0.418*** (0.037)	0.248*** (0.062)
<i>C. SDG buckets</i>						
Climate	0.032	0.104*** (0.014)				
Environment	0.040	0.119*** (0.013)				
Job and Equity	0.079	0.157*** (0.017)				
Number of investors/ Deals	20,228/360,158	277/8,125	68/1,422	90/1,849	121/4,005	47/1,062

Specification: The unit of observation is VC or growth equity investors with an investment by May 2021. In column 1, we present the mean of the outcome shown in the rows for Traditional investors. In columns 2–6, we show the coefficient and standard error of the difference between Traditional investors and the group indicated in the header of the column. Robust standard errors are reported in parentheses and are not clustered as the data are cross-sectional at the investor level with no repeated observations.

Outcomes: Outcomes are described in the rows of the table and represent averages at the investor level. In panel A, we present summary statistics of the investors' portfolios. In panel B, we show what fraction of an investor's portfolio companies are headquartered across the global regions listed in the panel. Advanced economies includes Canada, the United Kingdom, North/South/Western Europe, the Middle East and North Africa, Oceania, and East Asia. Developing economies includes Eastern Europe, Russia, Central Asia, Latin America and the Caribbean, Southeast Asia, South Asia, and Sub-Saharan Africa. In panel C, we show the fraction of the investors' portfolio companies that are classified in the SDG buckets listed in the panel. The final row of the table shows the number of investors and number of deals that fall in each of the categories indicated in the column header.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: PitchBook data feed as of May 2021.

average investment in a financing round for traditional investors is \$8.7 million, as against \$5.0 million for impact investors. Median transaction sizes are much more similar.¹⁹

¹⁹ Like most data sources derived from securities filings, such as U.S. Securities and Exchange Commission Form D, PitchBook does not typically identify the amount of capital or ownership stake of each individual investor in each investment round, just the aggregate amount in the round. Thus, these estimates are inexact: we assume that the contributions of each financier in the round are equal.

In panel B of [Table 1](#), we also see significant differences in investment location. Relative to traditional investors, impact investors are more likely to invest in low income regions of the world (e.g., Sub-Saharan Africa, South Asia, Russia, Central Asia, and Latin America and the Caribbean) and less likely to invest in high-income places outside the United States (e.g., Canada, East Asia, Europe, and the Middle East).

We also see differences in investment focus in panel C. The impact investors are three times more likely to be invested in companies classified into Climate and Environment, and almost twice as likely to invest in ones with a Jobs and Equity focus, than traditional investors.²⁰

Our final set of descriptive comparisons corroborates the earlier findings (e.g., [Kovner and Lerner 2015](#); [Barber et al. 2021](#)) of the relatively lower investment success of impact investors. As data on cash flows and valuations are not available, we instead focus on the probability that portfolio companies go public or are acquired. We see in [Table 2](#) that, on average, impact portfolio companies are substantially less likely to go public or be acquired (a 2.7-percentage-point-lower probability, relative to a mean of 16.5% in traditional companies). These patterns hold if we focus on acquisitions or IPOs only. Finally, we condition on an exit occurring and report the average number of months to exit. Conditional on a successful exit, impact portfolio companies take 15 additional months to reach a successful exit.

2. Do Impact Investors Prioritize Impact?

To advance the debate about whether impact investors are differentiated in seeking social impact versus merely “impact washing,” we characterize the nonfinancial features of impact investments.

2.1 Alignment with the federal government

We examine the overlap between investors’ portfolio companies and the priorities of the U.S. federal government, which we argue below may be proxies for social impact. First, we examine whether companies received funds from the SBIR program (either a Phase I or Phase II award). The SBIR is the largest government program funding high-potential entrepreneurial firms and has been shown to be associated with greater firm innovativeness and substantial positive social spillovers ([Howell 2017](#); [Myers and Lanahan 2022](#)). The objectives of the SBIR program include to “meet Federal research and development needs” and to “increase private sector commercialization of innovations developed through Federal R&D

²⁰ In [Table C.1](#), we characterize both the sectoral and geographic distribution of portfolio companies of impact investors.

Table 2
Exit outcomes of impact investments

	(1) Traditional company mean	(2) Impact present	(3) Climate impact present	(4) Environment impact present	(5) Job and Equity impact present	(6) Concessionary impact present
IPO, merger, or acquisition	0.165	-0.027*** (0.005)	-0.012 (0.011)	-0.038*** (0.009)	-0.052*** (0.006)	-0.049*** (0.011)
Merger or acquisition	0.133	-0.017*** (0.004)	-0.012 (0.011)	-0.029*** (0.008)	-0.042*** (0.005)	-0.035*** (0.011)
IPO	0.032	-0.009*** (0.002)	0.000 (0.006)	-0.008** (0.004)	-0.011*** (0.002)	-0.014*** (0.004)
Months btwn first deal and exit	62.562	15.268*** (1.677)	17.012*** (3.789)	21.924*** (3.975)	17.814*** (2.790)	22.937*** (4.666)
Sector FE		Yes	Yes	Yes	Yes	Yes
First financing year FE		Yes	Yes	Yes	Yes	Yes
Number of impact firms	0	6,066	1,019	1,425	3,063	881
Number of traditional firms	204,640	204,640	204,640	204,640	204,640	204,640

Specification: The unit of observation is companies funded by a VC or growth equity investors by May 2021. In column 1, we present the mean of the outcome shown in the rows for Traditional companies. Traditional companies have never had an impact investor; there are 204,640 in our sample. In columns 2–6, we show the coefficient and standard error of the difference between Traditional companies and the group indicated in the header of the column. In column 2, Impact present indicates a company that has at least one impact investor; there are 6,066 such companies. Columns 3–5 present the difference relative to column 1 for impact companies that have received at least one investment from an investor specializing in the SDG bucket listed in the column header. Impact companies that have not received an investment from an investor specializing in the relevant SDG bucket are excluded from the respective column. Similarly, nonconcessionary impact investments are not included in the analysis for column 6. Fixed effects for sector and first year of financing are noted. Robust standard errors are reported in parentheses and are not clustered as the data are cross-sectional at the company level with no repeated observations.

Outcomes: Outcomes are described in the rows of the table. The outcome in row 1 is whether the company had an IPO, merger, or acquisition. It is the union of the outcomes in rows 2 and 3. The outcome in row 4 is the number of months between the date of the first deal and the date of an exit (IPO, acquisition, or merger). The sample in row 4 is limited to companies that achieve an exit (as defined above) and for which the first investment date and the exit date are not missing.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: PitchBook data feed as of May 2021. We use PitchBook’s detailed 215 industry classifications as sector fixed effects.

funding,” so we would anticipate a strong correlation between awards and the public’s research and innovation needs.²¹

Second, we examine a portfolio company’s patents with a noted government interest. As described in [Gross and Sampat \(2024\)](#), government agencies, grantees, and contractors have been required to identify all patents where the government had rights, including not simply those where the government agency was an assignee but also those that were government licensed or government funded, even if held by another entity. Again, entrepreneurial firms awarded such patents should be more likely to be working in areas of public interest. Both of these data sets are national and comprehensive (reporting all awards made), making them ideal to match to our data set, which contains the universe of impact and traditional portfolio companies.

²¹ See Small Business Innovation Development Act of 1982, Pub. L. No. 97-219.

These data also allow us to test for two channels through which impact investors could have an effect. First, they could select for investment companies that are aligned with the federal government. Such activity might not be “additional” if they simply crowd out other investors by offering similar quantities of capital on similar terms.²² Second, there could be a “treatment effect,” whereby companies that receive impact investment subsequently become more aligned with the government. This would constitute postinvestment additionality.

Table 3 tests for selection effects. We restrict the sample (here, as well as in Tables 4, 7, 11, and C. II) exclusively to U.S.-based firms, as these firms are much more likely to interact with the U.S. government.

We estimate, with one observation per company,

$$y_i = \alpha + \beta \text{ImpactPresent}_i + \gamma_i + \delta_i + \eta_i + \varepsilon_i, \quad (1)$$

where y_i is an indicator variable as to whether the company received an SBIR award or filed a government-interested patent, or the logarithm of the count of such events, in the 5 years prior to the first financing round; ImpactPresent_i is a dummy denoting whether an impact investor participated any funding round; and γ_i , δ_i , and η_i represent state, founding year, and industry fixed effects, respectively. We employ heteroskedasticity-robust standard errors.

Columns 1 and 2 show in each case that impact investors’ portfolio companies are substantially more likely to have received an SBIR award or to have filed for a government-interested patent. The estimates range from an 80 to a 180% higher probability of having at least one award or pending or granted patent: for instance, in columns 1 and 2 of panel A, the probability goes from 1.9 to 2.0%. We similarly see an increase in the number of such awards in columns 3 and 4. The effect is robust to industry, location, and time fixed effects, addressing some potential confounding factors.²³

Table 4 focuses instead on the treatment effects of impact investments. We look at the consequences of impact financing on the probability a company receives an SBIR award or applies for a government-interested patent.

²² Alternatively, this could be additional if impact investors offer better terms or if more dollars chasing publicly useful ideas entices more entrepreneurs to enter these sectors; however, we lack data to test this hypothesis.

²³ One might worry that firms in the portfolio of impact funds are more likely to be financially constrained, and that these constraints, rather than anything about the nature of the focus of their business, lead them to seek public funding. Thus, we create two alternative measures of public interest in columns 1 and 2 of Internet Appendix Table C.II, which are independent of a specific firm’s financial constraints. We first identify industries that are government-interest aligned, by ranking the 41 PitchBook industries by the percentage of PitchBook firms in that industry that receive an SBIR award from 1983 to 2022. We find that impact portfolio companies are more likely to be in industries that are in the top 20% (or top 30%) of government awards. Similarly, we rank four-digit Combined Patent Classification (CPC) system subclasses by the probability that a firm has a government-interested patent between 1963 and 2020. Again, we find that impact portfolio companies are substantially more likely to file patents in these specific subclasses.

Table 3
Do impact investors invest in government-interested companies? An analysis using SBIR awards and patents

	(1) Any predeal award	(2) Any predeal award	(3) log predeal awards	(4) log predeal awards
<i>A. SBIR awards</i>				
Impact present	0.020*** (0.003)	0.019*** (0.003)	0.027*** (0.005)	0.025*** (0.004)
State fixed effect	No	Yes	No	Yes
Founding year fixed effect	No	Yes	No	Yes
Industry fixed effect	No	Yes	No	Yes
Adj. R-squared	.001	.024	.001	.021
N observations	74,622	74,622	74,622	74,622
Mean traditional	0.011	0.011	0.013	0.013
	(1) Any predeal gov't-interested patent	(2) Any predeal gov't-interested patent	(3) log predeal gov't-interested patents	(4) log predeal gov't-interested patents
<i>B. Government-interested patents</i>				
Impact present	0.004** (0.002)	0.003* (0.002)	0.005** (0.002)	0.004* (0.002)
State fixed effect	No	Yes	No	Yes
Founding year fixed effect	No	Yes	No	Yes
Industry fixed effect	No	Yes	No	Yes
Adj. R-squared	.000	.013	.000	.011
N observations	74,622	74,622	74,622	74,622
Mean traditional	0.004	0.004	0.004	0.004

Specification: Both panels in this table estimate Specification 1 in the paper. The unit of observation is U.S. companies funded by VC or growth equity investors by May 2021. Impact present indicates whether the company received an investment from an impact investor. In panel A, the Mean traditional value under columns 1 and 2 is the proportion of traditional firms that have received a predeal SBIR award. The Mean traditional value under columns 3 and 4 is the average log of predeal SBIR awards among traditional firms. In panel B, the Mean traditional value under columns 1 and 2 is the proportion of traditional firms that have filed predeal for a government-interested patent. Mean traditional value under columns 3 and 4 is the average log of government-interested patents among traditional firms. Prior to taking the log of SBIR awards and patents, we add a value of one to account for zeros. Fixed effects for state, founding year, and industry are noted. Robust standard errors are reported in parentheses and are not clustered as the data are cross-sectional at the company level with no repeated observations.

Outcomes: In panel A, the outcome in Columns 1 and 2 is a dummy indicating if the firm received an SBIR award within 5 years prior to the first deal; in columns 3 and 4, it is the log of the number of awards within 5 years prior to the first deal. We count the number of SBIR awards of either phase and do not count rejected SBIR proposals. In panel B, the outcome in columns 1 and 2 is a dummy indicating if the firm filed any government-interested patent in the 5 years prior to the first deal; in columns 3 and 4, it is the log of the number of government-interested patents filed 5 years prior to the first deal. We count the number of government-interested patents ultimately granted, which excludes unsuccessful or withdrawn applications.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: PitchBook data feed as of May 2021. SBIR awards are from the SBIR database at www.sbir.gov/awards# (version of March 27, 2024). Government-interested patents are from the sample created by Gross and Sampat (2024).

Here we modify our approach in several ways. First, rather than having 1 observation per company as in Table 3, we employ up to 11 annual observations for each entity, from 5 years before to 5 years after the first investment (impact, or, for the matched sample, traditional investment). We do not include years before the company was founded or years after

2022 for the SBIR outcome, or years after 2020 for the patent outcome, even if they fall within the 11-year window. Second, seeking as similar matches as possible, we match each impact portfolio company to traditional companies along three dimensions: region (Northeast, Southeast, Midwest, and West), industry (the 41 PitchBook industry groups), deal type (we distinguish between first-round VC, later-round VC, and growth equity), and year. In the latter case, we require the non-impact-companies to have their earliest financing round in the year before, of, or after that of the earliest impact financing round of the impact portfolio company.²⁴

In each case, we examine the following difference-in-differences specification:

$$y_{it} = \alpha + \beta_1 \text{ImpactPresent}_i + \beta_2 \text{Raise}_t + \beta_3 \text{ImpactPresent}_i * \text{Raise}_t + \gamma_i + \delta_t + \varepsilon_{it}, \quad (2)$$

where y_{it} indicates if the company was awarded an SBIR award or filed for a government-issued patent in the year of the observation. ImpactPresent_i is a dummy denoting an impact-backed company, Raise_t is a dummy equal to 1 if the observation is from the year of the financing or the 5 years thereafter, and γ_i and δ_t denote firm and observation year fixed effects. Standard errors are clustered at the company level, as this is the unit at which our “treatment” (i.e., investor type) is assigned (Bertrand et al. 2004).

The difference-in-differences model represented by Specification 2 provides the causal estimate of the effect of raising impact capital (beyond that of raising traditional VC). The underlying assumption is that, for the outcomes of interest, had the impact-backed companies instead raised traditional VC, they would have evolved at the same rate as the matched companies that also raised traditional VC. In support of this assumption, Figure C.1 plots the event study analog of Specification 2. Reassuringly, it is visually apparent that prior to their fundraising event, the outcomes of sample companies that will eventually raise impact and traditional VC exhibit parallel trends. Parallel pretrends are confirmed in a regression framework in Table C.III.

Table 4 provides the estimates of Specification 2 and shows evidence of a treatment effect for the SBIR measure. Both companies financed by traditional venture investors and those financed by impact investors are more likely to receive an SBIR award after the financing event. But the increase is much larger for impact portfolio companies. For instance, examining column 2, we see that companies backed only by traditional

²⁴ For the SBIR analysis, we dropped firms if multiple firms had the same name as an SBIR-awarded company (9 firms) or if the SBIR award year was missing (1 firm). Thus, we have 2,686 impact companies and 49,939 traditional companies in the SBIR analysis and 2,686 impact companies and 49,949 traditional companies in the patent analysis.

Table 4
Do impact investors raise the likelihood of government interest in their portfolio companies?

	(1) Any SBIR award	(2) Any SBIR award	(3) Any gov't-interested patent	(4) Any gov't-interested patent
Impact * Raise	0.0091** (0.0036)	0.0068** (0.0031)	-0.0002 (0.0020)	-0.0004 (0.0019)
Impact	0.0196*** (0.0044)		0.0076*** (0.0019)	
Raise	0.0184*** (0.0007)	0.0072*** (0.0006)	0.0020*** (0.0003)	0.0006* (0.0003)
Firm fixed effect	No	Yes	No	Yes
Year fixed effect	No	Yes	No	Yes
Adj. R-squared	.0032	.7438	.0008	.4327
N observations	397,637	397,637	358,259	358,259
Mean value	0.0375	0.0375	0.0051	0.0051

Specification: This table estimates Specification 2 in the paper. The unit of observation is the company-year. The sample includes U.S. companies funded by impact investors by May 2021 and a matched set of those backed by VC or growth equity investors (see [Section 2.1](#) for details). In columns 1 and 2, there are 2,686 impact companies and 49,939 traditional companies. In columns 3 and 4, there are 2,686 impact companies and 49,949 traditional companies. Impact is a binary indicator for whether the company has at least one impact investor. Raise is a binary indicator that takes a value of one for the year in which the company has an initial impact fundraising or an initial VC or growth equity financing for matching firms (normalized to year 0 in our analysis) and all years after (years 1–5); it takes a value of 0 for all years prior to the raise (normalized to years –5 to –1). Impact*Raise is an interaction between these two binary indicators. Fixed effects for firm and year are noted. Robust standard errors clustered at the company level are in parentheses.

Outcomes: The outcome in columns 1 and 2 is a dummy indicating if the firm received an SBIR award in or by that year. We count SBIR awards from either phase and do not count rejected SBIR proposals. The outcome in columns 3 and 4 is a dummy indicating if the firm filed a government-interested patent in that year. We only examine government-interested patents ultimately granted, which excludes unsuccessful or withdrawn applications.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: PitchBook data feed as of May 2021. SBIR awards are from SBIR database at www.sbir.gov/awards# (version of March 27, 2024). Government-interested patents are from the sample created by [Gross and Sampat \(2024\)](#).

investors have a 0.7% increase in the likelihood of receiving an SBIR award postfinancing, while companies backed by an impact investor have a 1.4% increase. The difference is substantial in economic magnitude relative to the average probability of 3.8% across the sample. Columns 3 and 4 show there is no evidence of a similar treatment effect of impact investment for government-interested patents. These findings complement the results of [Howell \(2017\)](#), which uses a regression-discontinuity design to show that SBIR awards have a positive impact on innovation and on the probability firms receive venture funding. Our paper examines a related but distinct question, showing that the type of investor a firm attracts shapes its engagement with programs like the SBIR. This suggests a potentially virtuous cycle: impact investors induce firms to apply for SBIR grants, which in turn improve firm outcomes, with a combined effect that may increase the long-run success of firms aligned with public mission objectives.

2.2 Portfolio company headquarters

We next compare the headquarters locations of the portfolio companies, with an eye toward the demographic characteristics that might indicate a location is economically disadvantaged. In column 1 of [Table 5](#), we present the mean for the companies entirely funded by traditional investors. Column 2 presents coefficients from regression Specification 1 using a sample of all impact-backed-firms and non-impact-firms, with fixed effects for industry and the year of the first financing. Note that each cell in the table represents the coefficient of interest— β in Specification 1—from a distinct regression. We defer discussion of columns 3–6 until [Section 4](#).

Across countries, we see that impact investors' portfolio companies are headquartered in nations with lower per capita gross domestic product. When we compare across counties within the United States, however, the results are less clear. Impact investors' portfolio companies are more likely to be headquartered in counties with better educated residents and fewer Blacks and Hispanics. Deaths from drugs and alcohol as a percentage of the population are slightly higher for counties with impact portfolio companies, but the difference is modest in magnitude. As we will see in [Section 4](#), the aggregated differences of impact and non-impact-portfolio companies mask substantial heterogeneity, which we uncover by looking across distinct categories of impact investors.

3. Impact Investors and Traditional Investors

We next seek to understand the ways in which impact and traditional investors interact. Several theories of impact investing suggest that impact investors can create more impact if they differentiate themselves from traditional venture investors by supporting high-impact companies that might not have otherwise been able to raise capital (e.g., [Brest and Born 2013](#); [Oehmke and Opp 2020](#); [Green and Roth forthcoming](#)). Put another way, these theories predict that impact investors that are “additional” have more impact.

To shed light on the degree to which impact investors differentiate themselves, we focus on three dimensions: the frequency of syndicated investments with a traditional investor, the degree to which impact investors specialize in the financing of new industries, and the extent to which impact investors provide funding during market downturns.

3.1 Do impact investors coinvest with traditional investors?

[Table 6](#) reports the frequency with which impact investors coinvest with a traditional investor in the same firm and financing round. We include all financing rounds in the sample (in some cases more than one per portfolio firm). Column 1 looks at financing rounds involving exclusively traditional investors. Transactions are coded as one if more than one

Table 5
What are the socioeconomic predictors of impact investments?

	(1) Traditional company mean	(2) Impact present	(3) Climate impact present	(4) Environment impact present	(5) Job and Equity impact present	(6) Concessionary impact present
U.S.-based companies						
Median household income USD	70,000.34	-396.81 (317.92)	4,089.29*** (762.10)	3,230.64*** (737.27)	-282.56 (505.48)	410.98 (943.43)
Population density (person/sq.mi)	9,519.68	-1.59 (372.31)	795.45 (798.72)	-1,327.65* (731.34)	1,609.51** (682.44)	-956.39 (1,042.03)
Black and Hispanic population percent * 100	30.12	-1.30*** (0.34)	-0.78 (0.74)	-2.02*** (0.75)	1.87*** (0.54)	-0.95 (0.96)
Deaths from drugs or alcohol percent * 100	0.69	0.01** (0.00)	-0.02*** (0.01)	-0.03*** (0.01)	-0.00 (0.00)	0.03*** (0.01)
No high school diploma percent * 100	13.99	-0.33*** (0.10)	-0.16 (0.25)	-0.61** (0.25)	0.08 (0.15)	-0.55* (0.29)
Bachelor or graduate degree percent * 100	48.16	1.16*** (0.24)	3.39*** (0.58)	2.57*** (0.57)	1.01*** (0.35)	0.70 (0.71)
Poverty percent * 100	13.11	-0.02 (0.05)	-0.15 (0.11)	-0.30*** (0.11)	0.16** (0.08)	0.02 (0.14)
All companies						
GDP per capita USD	40,750.87	-4,123.75*** (330.85)	706.49 (713.52)	-12,467.64*** (731.65)	-9,220.20*** (493.83)	-7,432.12*** (858.29)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
First financing year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of impact firms	0	6,066	1,019	1,425	3,063	881
Number of traditional firms	204,640	204,640	204,640	204,640	204,640	204,640

Specification: The unit of observations is companies funded by VC or growth equity investors by May 2021. In column 1, we present the mean of the outcome shown in the rows for Traditional companies. Traditional companies have never had an impact investor; there are 204,640 in our sample. In columns 2–6, we show the coefficient and standard error of the difference between Traditional companies and the group indicated in the column header. In column 2, Impact present indicates a company that has at least one impact investor; there are 6,066 such companies. Columns 3–5 present the difference relative to column 1 for impact companies that have received at least one investment from an investor specializing in the SDG bucket listed in the column header. Impact companies that have not received an investment from an investor specializing in the relevant SDG bucket are excluded from the respective column. Similarly, nonconcessionary impact investments are not included in the analysis for column 6. Fixed effects for industry and first year of financing are noted. Robust standard errors are reported in parentheses and are not clustered as the data are cross-sectional at the company level with no repeated observations.

Outcomes: Outcomes are described in the rows of the table. The outcomes in rows 1–6 are calculated at the U.S. county level and hence only U.S. companies are considered. In row 7, GDP per capita (USD) is presented at the country level. Outcomes are assigned to companies based on the headquarters of the company as specified in the first round of investment. Observation numbers vary across rows due to availability of information on firm location or outcome data.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: PitchBook data feed as of May 2021. Household income (*United States Census Bureau 2011*), population density (*United States Census Bureau 2021*), education attainment and Black-Hispanic population (*United States Census Bureau 2020*), causes of drug and alcohol deaths (*National Center for Health Statistics 2020*), and GDP per capita (constant 2015 [USD]) (*World Bank 2020*).

traditional investor invested in the round and zero otherwise. The second column conditions on there being at least one impact investor in the round. Transactions are coded as one if there is at least one traditional investor in addition to an impact investor in the round and zero otherwise.

Table 6
Percentage of rounds that are coinvested with a traditional investor

	(1) Traditional coinvests with traditional	(2) Impact coinvests with traditional	(3) Climate coinvests with traditional	(4) Environmental coinvests with traditional	(5) Job and Equity coinvests with traditional	(6) Concessionary coinvests with traditional
Entire sample	31.70%	60.27%	73.49%	58.79%	54.01%	64.41%
VC rounds	33.44%	63.65%	79.41%	64.72%	57.22%	67.04%
PE growth rounds	11.75%	26.13%	29.59%	24.54%	25.97%	27.14%
2000–2005	56.81%	65.85%	86.27%	55.32%	52.00%	50.00%
2005–2010	45.38%	68.44%	71.50%	61.31%	63.01%	45.68%
2010–2015	33.84%	56.03%	73.73%	54.79%	44.93%	63.12%
2015–2020	31.71%	63.54%	77.45%	61.25%	56.91%	69.48%
2020–2022	42.00%	77.83%	81.92%	75.78%	78.50%	91.89%
1st round	24.39%	45.35%	61.51%	41.45%	38.36%	51.33%
Later round	58.28%	78.00%	84.82%	79.56%	71.83%	79.55%
Total deals	360,158	8,125	1,422	1,849	4,005	1,062

In this table, we present the percentage of financing rounds that are coinvested with a traditional investor. In column 1, we limit the sample to financing rounds that only have traditional investors (and so the numbers reflect the percentage of rounds with more than one traditional investor). In column 2, the sample comprises all financing rounds with at least one impact investor. Columns 3–6 show the percentages for financing rounds that include at least one Climate, Environment, Job and Equity, and concessionary investor, respectively. At the bottom of the table are the total rounds in each of the relevant samples. Row 1 presents the coinvestment percentages for the full sample reflected in the column headers. Rows 2 and 3 present the coinvestment percentages for the samples restricted to either VC or growth equity. Rows 4 through 8 present the coinvestment percentages for the samples restricted by financing year. Rows 9 and 10 present the coinvestment statistics for the sample restricted by whether the round is the first round in which a traditional/impact/SDG impact/concessionary impact investor was present or whether it is a subsequent round.

Source: PitchBook data feed as of May 2021.

The analysis suggests that impact investors, far from being fully segmented from traditional investors, are more likely to share transactions with traditional funds than traditional investors are to share transactions with one another. At 60%, the probability of coinvesting with a traditional investor is almost twice as high for the impact-present rounds as it is for the rounds where no impact investors are present.²⁵ These patterns are also seen when we look at VC rounds exclusively (where the probability of coinvestment is greater in general). Coinvestment is also substantially higher among impact investors in growth equity rounds compared to traditional investors. Of course, the extent of coinvestment we observe may reflect the small average investment size of impact investors noted in Table 1, which may be a consequence of modest fund sizes.

3.2 Do impact investors influence traditional investors?

While the majority of impact investments are coinvested with traditional investors, impact investors may still be additional if they influence the

²⁵ Note that here we focus on coinvestment in the same financing round. Analyses that look at how many investors coinvest over the life of the portfolio company find higher rates of coinvestment (see, e.g., Nanda and Rhodes-Kropf [2019]).

investment activity of their coinvestors. To investigate this possibility, we examine the behavior of traditional investors that have coinvested with an impact investor that subsequently raises a new fund. Given an increased rate of investment activity in the years following a fund raise, if impact investors influence their traditional coinvestors, we should see that the deals of these traditional investors should be more similar to impact deals following the impact fund raise.

Specifically, we utilize an event-study framework and estimate the following model:

$$y_{ijt} = \alpha + \beta \text{Fundraise}_{jt} + \gamma_j + \delta_t + \epsilon_{ijt}. \quad (3)$$

Here, the sample is all investments made into a portfolio company i by traditional private investor j in year t for all i, j, t in our data. Fundraise_{jt} is a dummy variable taking a value of 1 if (1) an impact investor has raised a fund in the last 1, 2, or 3 years before year t , depending on the specification and (2) that impact investor coinvested with traditional investor j in any of the 3 years preceding the fundraising. γ_j is an investor fixed effect, and δ_t is a fixed effect of the year of the transaction. We also present an alternative specification that controls for the number of coinvestments each traditional investor has made with an impact investor. Standard errors are clustered at the investor level, as this is the unit at which our “treatment” (i.e., an impact coinvestor’s fundraising event) is assigned (Bertrand et al. 2004).

We seek to understand whether traditional investor j ’s investments more closely resemble impact investments in the years following the fundraising event of an impact investor with whom they have previously coinvested. We operationalize resemblance to impact investments along two dimensions. First, as in Section 2.1, we measure portfolio company alignment with government priorities, as proxied by having an SBIR award or government-interested patent. Second, we measure whether the portfolio company is a “pioneer” in a nascent industry—that is, among the first 10, 20, 30, or 40 companies financed in its PitchBook industry in our data. As we document in Section 3.3, impact investors are systematically more likely than traditional investors to fund such pioneers. If impact investors have a causal influence on the investment activities of their prior traditional coinvestors, β will be positive. This identification strategy builds upon the literature studying the transmission of shocks across firms and investors in a network (e.g., Khwaja et al. 2011; Townsend 2015; Nanda and Rhodes-Kropf 2019).²⁶

²⁶ Perhaps closest to our strategy is Khwaja et al. (2011), which studies a firm network where links between firms are defined by having a director in common. That paper finds that when a firm i is linked to another firm j that experiences a positive shock, firm i ’s probability of raising external financing rises sharply. Similarly, we study the outcomes of a traditional investor i when an impact investor j to which i is linked experiences the positive shock of a fundraising event.

Table 7
Are impact investors guiding traditional venture investors to new deals?

	Any SBIR award		Any gov't-interested patent	
	(1)	(2)	(3)	(4)
<i>A. Fundraise 0–1 years prior</i>				
Impact fundraise	0.006** (0.002) [0.002,0.010]	0.005* (0.002) [0.001,0.009]	0.004** (0.001) [0.001,0.006]	0.003** (0.001) [0.001,0.005]
Control for # impact coinvestments	No	Yes	No	Yes
Investor FE	Yes	Yes	Yes	Yes
Deal year FE	Yes	Yes	Yes	Yes
Adj. R-squared	.091	.091	.071	.071
Mean traditional	0.068	0.068	0.023	0.023
<i>B. Fundraise 0–2 years prior</i>				
Impact fundraise	0.004* (0.002) [0.000,0.008]	0.003 (0.002) [−0.001,0.007]	0.004** (0.001) [0.001,0.006]	0.003** (0.001) [0.001,0.006]
Control for # impact coinvestments	No	Yes	No	Yes
Investor FE	Yes	Yes	Yes	Yes
Deal year FE	Yes	Yes	Yes	Yes
Adj. R-squared	.091	.091	.071	.071
Mean traditional	0.068	0.068	0.023	0.023
<i>C. Fundraise 0–3 years prior</i>				
Impact fundraise	0.003 (0.002) [−0.001,0.008]	0.002 (0.002) [−0.002,0.006]	0.004** (0.001) [0.001,0.006]	0.003* (0.001) [0.001,0.006]
Control for # impact coinvestments	No	Yes	No	Yes
Investor FE	Yes	Yes	Yes	Yes
Deal year FE	Yes	Yes	Yes	Yes
Adj. R-squared	.091	.091	.071	.071
Mean traditional	0.068	0.068	0.023	0.023
N observations	267,142	267,142	267,142	267,142

Specification: This table estimates Specification 3 in the paper. The unit of observation is the investor-company-round. The sample is limited to U.S.-based companies of traditional investors. In columns 2 and 4, we introduce a control for the total number of coinvested deals prior to the deal year. In each panel, the independent variable equals 1 if a prior impact coinvestor—with the most recent coinvestment in 3 years prior to the deal—has raised a new fund within the time frame indicated by the panel title (Fundraise 0–1 years prior, Fundraise 0–2 years prior, or Fundraise 0–3 years prior), including the deal year. Mean traditional reports the proportion of traditional investors' deals in firms with an SBIR award (columns 1 and 2) or a government-interested patent (columns 3 and 4). Robust standard errors, clustered at the investor level, are in parentheses. 95% confidence intervals are reported in brackets.

Outcomes: In columns 1 and 2, Any SBIR award takes a value of 1 if the firm has received an SBIR award and 0 otherwise. We consider SBIR awards from either phase and do not count rejected SBIR proposals. In columns 3 and 4, Any government-interested patent takes a value of 1 if the firm has filed a government-interested patent and 0 otherwise. We consider only government-interested patents ultimately granted, which excludes unsuccessful or withdrawn applications.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: PitchBook data feed as of May 2021. SBIR awards are from SBIR database at www.sbir.gov/awards# (version of March 27, 2024). Government-interested patents are from the sample created by Gross and Sampat (2024).

The results for portfolio companies' alignment with federal priorities are presented in Table 7, where each cell corresponds to a distinct regression, with the outcome variable represented in the column and the number in the cell corresponding to the estimate of β from Specification 3. For both outcomes, we see that the portfolio companies of traditional investors are

more likely to be aligned with government priorities following their impact coinvestor's fundraising event. For instance, in panel A we see that the portfolio companies that receive financing from traditional investors after the focal fundraising event are 7.4 to 8.8% more likely to have received an SBIR award, and 13.0 to 17.4% more likely to have a government-interested patent. The estimate for SBIR awards becomes somewhat weaker in panels B and C, as we broaden the definition of $Fundraise_{jt}$ to include more years, but remains statistically significant in panel B, and the estimate for government-interested patents is stable across the panels.

Table 8 reports analogous estimates with the four pioneer measures as outcome variables. The results are broadly consistent with those for government priority alignment in Table 7. The estimated effects are most consistent for the Pioneer (First 20) measure: across all three panels, the deals of traditional investors are about 0.3 percentage points—roughly 10% relative to the traditional baseline of 3.1%—more likely to involve a pioneering firm in the years following an impact coinvestor's fundraising event. Estimates for the Pioneer (First 30) measure are similar in magnitude and marginally significant, while coefficients for the most narrow (First 10) and most expansive (First 40) thresholds are positive but smaller and less precisely estimated. Taken together with Table 7, these patterns are consistent with impact investors influencing the deal flow of their prior coinvestors.

At least two mechanisms could drive this influence. First, impact investors may identify promising companies as the lead investor and bring traditional coinvestors to these deals. Second, impact investors may have indirect influence on their traditional coinvestors through discussions about promising sectors as well as demonstration effects whereby their coinvestors observe and emulate an impact investor's recent deals. We can confidently rule out the first of these mechanisms. Specifically, we reestimate Specification 3, where the outcome variable is whether traditional investor j had an impact coinvestor in deal i . The estimates, presented in Table C. IV, suggest that traditional investors are actually 0.06 percentage points less likely to coinvest with an impact investor following a fundraising event, consistent with the possibility that impact investors that have recently raised a fund have more capital and are less in need of coinvestors. This suggests that indirect influence is the operative channel for our results.

3.3 Do impact investors finance novel industries?

Impact investors often argue that part of their strategy is to support companies in novel markets and industries that carry substantial risk that may deter traditional investors. Successful investments may attract traditional investors to the category. For instance, many early debt and equity impact investing funds were created to finance companies in the

Table 8
Are impact investors guiding traditional venture investors to new deals? Pioneer analysis

[illegible]

Specification: This table estimates Specification 3 in the paper. The unit of observation is the investor-company-round. The sample is limited to U.S.-based companies of traditional investors. In columns 2, 4, 6, and 8, we introduce a control for the total number of coinvested deals prior to the deal year. In each panel, the independent variable equals 1 if a prior impact coinvestor—with the most recent coinvestment in 3 years prior to the deal—has raised a new fund within the time frame indicated by the panel title (Fundraise 0–1 years prior, Fundraise 0–2 years prior, or Fundraise 0–3 years prior), including the deal year. Robust standard errors, clustered at the investor level, are in parentheses. 95% confidence intervals are reported in brackets.

Outcomes: The Pioneer outcomes are indicators for whether the company is among the first 10, 20, 30, or 40 companies within its industry to have a financing round in our data set. We use the first deal date of the indicator. If the first deal date is missing, we omit the company from the analysis. Mean traditional reports the proportion of traditional investors' deals in the pioneer firms indicated in the column headers.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PitchBook data feed as of May 2021. We use PitchBook's detailed 215 industry classifications of the companies in our sample.

newly emerging sector of microfinance, which focuses on making small loans to poor women in developing countries. The sector subsequently achieved substantial interest from traditional funds and ultimately many IPOs. The early support of impact investors might allow companies and industries to develop the business models with the demonstrated profitability necessary to attract traditional investors. In this analysis, we investigate this claim by measuring whether impact investors, relative to traditional investors, are more likely to support companies in nascent industries, using the 215 PitchBook-identified sectors.

Table 9 examines whether impact investors help create new industries by comparing their portfolio companies to portfolio companies with only traditional investors. It presents regression results using the specification

$$y_i = \alpha + \beta \text{ImpactPresent}_i + \delta_i + \varepsilon_i, \quad (4)$$

where y_i is a dummy denoting whether the company was a “pioneer,” using the four thresholds introduced in Section 3.2. (Rows 1–4 correspond to the first 10, 20, 30, and 40 companies financed within the firm’s PitchBook industry in our data.) As above, ImpactPresent_i is a dummy denoting whether an impact investor participated in any funding round. δ_i is a year-of-first-financing fixed effect. We compute robust standard errors.²⁷

We see that relative to traditional-only companies, impact-present companies are more likely to be pioneers, regardless of the definition of an industry pioneer.²⁸ In all cases, the probability of an investment by an impact fund being pioneering is roughly 50% higher than that of traditional investing.

3.4 Do impact investors finance firms in market downturns?

Another way impact investors could differentiate themselves is to provide funding when funding is relatively scarce. VC funding is intensely cyclical

²⁷ Table C.V provides corresponding estimates for the specification without year-of-first-financing fixed effects. The results are similar.

²⁸ The reader may be curious about the industries where the impact investors were especially well-represented. We find in an unreported analysis that the five PitchBook industries in which impact investors are most likely to support pioneers are Alternative Energy Equipment, Forestry Development/Harvesting, Horticulture, Other Utilities (largely composed of clean energy companies), and Plant Textiles. Within each of these industries, impact investors are present in between 20 and 30% of the pioneering deals. For concreteness, the following are examples of impact-backed pioneer companies in each of the aforementioned industries. Alternative Energy: Capstone Green Energy was incorporated in 1988 as a California based gas turbine manufacturer that specializes in microturbine power along with heating and cooling cogeneration systems. Forestry: Triton Timber was founded in 2000 in Victoria, Canada to develop technology to responsibly harvest the flooded and abandoned forests in reservoirs around the world. Horticulture: Nalweyo Seed Company Ltd (NASECO) was formed in 1996 and breeds, produces, and sells a variety of hybrid field crops and vegetables to local and international nongovernmental organizations, distributors, and smallholder farmers in Uganda and beyond. Other Utilities: Cogelec Energy was created in 2014 to provide energy for productive use and act as a catalyst for economic advancement in communities across Africa. Plant Textiles: AlgaLife, a Berlin and Israel-based startup established in 2016, seeks to develop algae-based materials for the fashion and textile industries.

Table 9
Do impact investors help create new industries?

	(1)	(2)	(3)	(4)	(5)	(6)
	Traditional company mean	Impact present	Climate impact present	Environment impact present	Job and Equity impact present	Concessionary impact present
Pioneer (first 10)	0.013	0.006*** (0.001)	0.010*** (0.003)	0.010*** (0.003)	0.007*** (0.002)	0.007* (0.003)
Pioneer (first 20)	0.026	0.012*** (0.002)	0.020*** (0.004)	0.029*** (0.004)	0.012*** (0.003)	0.018*** (0.005)
Pioneer (first 30)	0.038	0.019*** (0.002)	0.034*** (0.005)	0.045*** (0.005)	0.020*** (0.003)	0.026*** (0.006)
Pioneer (first 40)	0.050	0.024*** (0.003)	0.044*** (0.006)	0.060*** (0.005)	0.023*** (0.003)	0.037*** (0.007)
First financing year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of impact firms	0	6,066	1,019	1,425	3,063	881
Number of traditional firms	204,640	204,640	204,640	204,640	204,640	204,640

Specification: This table estimates Specification 4 in the paper. The unit of observation is companies funded by VC or growth equity investors by May 2021. In column 1, we present the mean of the outcome shown in the rows for Traditional companies. Traditional companies have never had an impact investor; there are 204,640 in our sample. In columns 2–6, we show the coefficient and standard error of the difference between Traditional companies and the group indicated in the column header. In column 2, Impact present indicates a company that has at least one impact investor; there are 6,066 such companies. Columns 3–5 present the difference relative to column 1 for impact companies that have received at least one investment from an investor specializing in the SDG bucket listed in the column header. Impact companies that have not received an investment from an investor specializing in the relevant SDG bucket are excluded from the respective column. Similarly, nonconcessionary impact investments are not included in the analysis for column 6. Fixed effects for first year of financing are noted. Robust standard errors are reported in parentheses and are not clustered as the data are cross-sectional at the company level with no repeated observations.

Outcomes: Outcomes are described in the rows of the table. The Pioneer outcomes are indicators for whether the company is among the first 10, 20, 30, or 40 companies within its industry to have a financing round in our data set. We use the first deal date of the company to create the indicator. If the first deal date is missing, we omit the company from the analysis.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: PitchBook data feed as of May 2021. We use PitchBook's detailed 215 industry classifications of the companies in our sample.

(Gompers and Lerner 2000; Inderst and Müller 2004; Gompers et al. 2008; Nanda and Rhodes-Kropf 2013), as is the financing of innovative firms more generally (Brown et al. 2009; Kerr and Nanda 2015; Haddad et al. 2022). Impact investors, motivated by social impact as well as financial returns, might therefore be expected to fill in some of the gap left by traditional venture investors in market downturns.

To examine this, we use as observations each distinct year t and industry sector s , with separate observations for impact and traditional investors i . We estimate

$$y_{sti} = \alpha + \beta_1 \text{Impact}_{sti} + \beta_2 \text{MarketBook}_{st} + \beta_3 \text{Impact}_{sti} * \text{MarketBook}_{st} + \gamma_s + \varepsilon_{sti}, \quad (5)$$

where y_{sti} is one of several measures of activity in that sector-year-fund type triple (e.g., the count of transactions or log dollar volume of transactions), and MarketBook_{st} is the lagged market-to-book ratio of

firms in that sector-year. We include sector fixed effects, and standard errors are clustered by sector and year as this is the level at which $MarketBook_{st}$ varies (Bertrand et al. 2004).

Our empirical approach requires data on both privately held and publicly traded firms in the same sector. Consequentially, rather than using the 215 PitchBook sectors, we use a broader measure. In particular, S&P assigns both the private firms in PitchBook and the public firms in Compustat to its Global Industry Classification Standard (GICS). We use the 11 broad sectors in the GICS: Communication Services, Consumer Discretionary, Consumer Staples, Energy, Financials, Health Care, Industrials, Information Technology, Materials, Real Estate, and Utilities. The specification includes fixed effects for each sector and year.

We calculate the market-to-book ratio for each sector following Daniel and Titman (2006), as described in Internet Appendix A.5.

Table 10 shows that, consistent with the findings in Gompers et al. (2008), there is a strong cyclicalitv among traditional venture investments across sectors. In contrast, cyclicalitv is essentially zero, and in some cases negative, for the impact investment transactions, indicating that impact investors do partially fill in the void left by traditional investors in market downturns. The results are robust when we use the contemporaneous, rather than lagged, market-to-book ratio (Table C. IV.)

In short, we see some striking patterns in the way that impact investors engage with traditional investors. While they are far more likely to syndicate deals with such investors, they are also quite differentiated in their behavior in some respects. In particular, they are more likely to invest in pioneering sectors and to resist public market signals.

4. Heterogeneity Among Impact Investors

Finally, we examine the behavior of impact investors with a focus on specific SDG sectors, as well as self-identified concessionary investors, with an eye to key differences. As described above, we classify impact investors with more than 20% of their transactions as being in a given “SDG bucket.”

Tables 1, 2, 5, 6, 9, and 10 separately analyze each subset of these impact investors in the final four columns of each table. Table 11 presents the analysis in the first column of Table 3 for various subsets of impact investors.

We see that in many respects, the activities of the various subclasses of impact investors are consistent. Table 6 highlights that all the subclasses of impact investors—investors focusing on the three SDG buckets and concessionary funds—are substantially more likely to coinvest with traditional investors than other investors. Similarly, Table 10 indicates that each subclass of impact funds avoids the strong cyclicalitv seen in traditional funds.

Table 10
Are impact investors cyclical investors?

	log deal number						
	(1) log deal number	(2) Deal number	(3) log deal amount	(4) Climate impact present	(5) Environment impact present	(6) Job and Equity impact present	(7) Concessionary impact present
Impact* MB _(t-1)	-0.232** (0.101)	-475.317* (235.460)	-0.150 (0.108)	-0.321** (0.143)	-0.303* (0.147)	-0.333** (0.123)	-0.369** (0.143)
MB _(t-1)	0.460*** (0.142)	442.641* (224.881)	0.602** (0.234)	0.471*** (0.143)	0.469*** (0.144)	0.477*** (0.140)	0.469*** (0.138)
Impact	-2.607*** (0.442)	170.777 (574.894)	-4.171*** (0.516)	-3.346*** (0.643)	-3.282*** (0.625)	-2.825*** (0.585)	-3.333*** (0.671)
Deal year fixed effect	No	No	No	No	No	No	No
Sector fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-squared	.648	.308	.577	.764	.751	.700	.779
N observations	682	682	682	682	682	682	682

Specification: This table estimates Specification 5 in the paper. The unit of observation is at a sector (based on the 11 GICS industry sectors)-by-year level for the period from 1990 to 2021 as well as impact versus traditional. Impact is a dummy that is equal to 1 for the deals that receive funding from at least one impact investor and 0 otherwise. MB_(t-1) is the lagged sector market-to-book ratio, calculated following [Daniel and Titman \(2006\)](#) and [Fama and French \(1992\)](#). All regressions include sector fixed effects. Robust standard errors are clustered by sector and year to account for industry trends and macroannual shocks and are reported in parentheses. Columns 4–7 are estimated using subsets of the impact companies in column 1, with each column using data calculated on all traditional investors, and the impact investors whose specialization is given in the column header.

Outcomes: The outcomes in columns 1 and 4–7 are the log of the total annual number of deals. The outcome in column 2 is the (unlogged) total annual number of deals. The outcome in column 3 is the log of the total annual deal amount.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: PitchBook data feed as of May 2021. The MB ratios are computed using Wharton Research Data Services.

On the other hand, we also observe apparently distinctive strategies by fund type. One of these is highlighted in [Table 5](#). There is a sharp differentiation in the parts of the United States selected for investments between the funds in the Climate and Environmental SDG buckets on the one hand and those focused on Jobs and Equity on the other. Portfolio companies of the former are headquartered in wealthier counties as measured by median household income; portfolio companies of the latter are headquartered in regions with 0.16% higher poverty rates. Similarly, the portfolio companies of Jobs and Equity firms are headquartered in areas with a substantial Black and Hispanic population and a higher population density. Overall, it seems that impact investors specializing in Jobs and Equity focus on economically disadvantaged areas within the United States, while the opposite is true for investors specializing in Climate and Environment.²⁹

²⁹ Another pattern of note is the differentiation between funds falling into the Climate and Environmental buckets internationally. Climate companies are concerned with reducing carbon footprint (e.g., renewable energy) and enhancing resilience against climate change, while Environment companies address a broader range of topics, including water conservation, land restoration, and sustainable farming, which aim at

Table 11
Do impact investors invest in government-interested companies? An analysis using SBIR awards and patents

	Any predeal award				
	(1) All impact present	(2) Climate impact present	(3) Environment impact present	(4) Job and Equity impact present	(5) Concessionary impact present
<i>A. SBIR awards</i>					
Impact present	0.020*** (0.003)	0.039*** (0.010)	0.035*** (0.009)	0.010** (0.004)	0.032*** (0.011)
Adj. <i>R</i> -squared	.001	.001	.001	.000	.000
N observations	74,622	72,146	72,145	72,807	71,989
Number of impact firms	2,981	505	504	1,166	348
Number of traditional firms	71,641	71,641	71,641	71,641	71,641
	Any predeal gov't-interested patent				
	(1) All impact present	(2) Climate impact present	(3) Environment impact present	(4) Job and Equity impact present	(5) Concessionary impact present
<i>B. Gov't-interested patents</i>					
Impact present	0.004** (0.002)	0.010* (0.005)	0.014** (0.006)	−0.000 (0.002)	0.002 (0.004)
Adj. <i>R</i> -squared	.000	.000	.000	−.000	−.000
N observations	74,622	72,146	72,145	72,807	71,989
Number of impact firms	2,981	505	504	1,166	348
Number of traditional firms	71,641	71,641	71,641	71,641	71,641

Specification: Both panels in this table estimate Specification 1 in the paper. The unit of observation is U.S. companies funded by VC or growth equity investors by May 2021. Impact present indicates whether the company received an investment from an impact investor. In both panels, columns 2–5 are estimated using subsets of the impact companies in column 1, with each column using data calculated on all traditional investors, and the impact investors whose specialization is given in the column header. Robust standard errors are reported in parentheses and are not clustered as the data are cross-sectional at the company level with no repeated observations.

Outcomes: The outcome in panel A is a dummy indicating if the firm received an SBIR award in the 5 years prior to the first deal. The outcome in panel B is a dummy indicating if the firm filed for a government-interested patent in the 5 years prior to the first deal. We only consider granted government-interested patents and successfully awarded SBIR grants.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: PitchBook data feed as of May 2021. SBIR awards are from the SBIR database at www.sbir.gov/awards# (version of March 27, 2024). Government-interested patents are from the sample created by Gross and Sampat (2024).

Correspondingly, when looking at the likelihood that portfolio companies have a successful exit—IPO, merger, or acquisition—in Table 2, we see the negative relationship between impact investment and successful exits is most concentrated among impact investors focused on Jobs and Equity and concessionary impact investors, and, to a lesser degree, among Environment impact investors. Impact investors focused on the Climate,

improving the condition of the environment today. Environment impact firms are consequently located more often in poorer countries compared to Climate firms.

on the other hand, exhibit higher rates of successful exits, which are statistically indistinguishable from the rates of exit for traditional investors.

Yet another set of differences appear in [Table 11](#). When we look at SBIR awards in panel A, we see that the portfolio companies of all types of impact investors have won such awards. When we look at government-interested patents in panel B, significant effects are confined to the portfolio companies of Climate and Environment investors. The coefficients for companies in the Jobs and Equity SDG bucket and concessionary funds are much smaller and insignificant. One possibility, suggested by the company profiles in [Internet Appendix B](#), is that these two subclasses of impact investors may primarily invest in firms for which formal intellectual property protection is less important.

Finally, [Table 9](#) highlights some differences with regard to how much impact investors specialize in nascent industries. While the portfolio companies of all subclasses of impact investors are more likely to be pioneers, these patterns are most pronounced for impact investors with a focus in Climate and Environment. As with the case of patents above, this result may reflect the innovative nature of Climate- and Environment-focused impact investors and portfolio companies.

5. Discussion

This paper analyzes the first comprehensive data set matching impact investors to their portfolio companies. We shed light on several long-standing debates about whether impact investors truly prioritize impact versus whether they are merely “impact washing” and the degree to which impact investors provide capital to firms and industries that would have been overlooked by traditional venture investors (i.e., the degree to which impact investors are additional, à la [Brest and Born \[2013\]](#)).

First, we analyze the nonfinancial determinants of impact investment. Relative to traditional investors, impact investors are more aligned with government (and presumably the public’s) interest, as measured by the rate at which their portfolio companies receive SBIR grants and file for government-interested patents. We find not only that impact investors are more likely to select companies that have received an SBIR award or filed a government-interested patent preinvestment, but also that postfinancing, impact investors seem to causally increase the likelihood that a company will receive an SBIR award. Relative to traditional investors, impact investors on average invest in countries with lower GDP per capita. Within the United States, there are fewer differences in the regions focused on by impact and traditional investors, but this result masks important heterogeneity across the impact investors.

Second, we analyze the degree to which impact investors substitute for or complement traditional venture investors. We document a strong degree

of coinvestment between traditional and impact investors, with upwards of 60% of all impact investments being coinvested with traditional investors. Nevertheless, we identify two important ways in which impact investors differentiate themselves from traditional investors, thereby expanding the financing frontier. Impact portfolio companies are more likely to be pioneers in their industries; this may serve to derisk the economics of unproven business models. And impact investors are more likely to provide capital during market downturns, which may add resilience to the venture financing landscape. We also find evidence that impact investors influence the investment activity of their traditional coinvestors.

Finally, we investigate heterogeneity in the above patterns across the impact investors. The portfolio companies of Jobs and Equity impact investors operate in substantially poorer and more disadvantaged counties within the United States, while the opposite is true for Climate and Environment impact investors. The portfolio companies of Climate and Environment investors, on the other hand, are more likely to be industry pioneers and to hold government-interested patents.

Our analysis suggests that impact investors go far beyond impact washing and provide capital to firms that would be overlooked by traditional investors. These results raise the question of how best to quantify and aggregate the social and financial tradeoffs associated with these investors. To what extent do the net societal benefits from impact investors' portfolio companies offset the lower financial returns to their limited partners documented in the earlier literature? How do the costs and benefits differ across different classes of impact groups? We hope that future research will help quantify these trade-offs.

Code and Data Availability. The replication code and data are available in the Harvard Dataverse at <https://doi.org/10.7910/DVN/4BWWE>.

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